Supplementary methods

Waiting time reverses re-pressurization yield rankings in polymer hydrogen-storage liners

S1. Independent verification of the constitutive update

The generalized Maxwell–Drucker–Prager update was verified at material-point and finite-element levels. Twenty-four single- and two-branch cases covered elastic response, return to the smooth yield cone, and return to its apex. The stress derivative was compared with a finite-difference tangent, and the condensed incremental potential was differentiated independently. The maximum relative discrepancies were 2.4111×10^{-9} for the tangent and 9.9979×10^{-10} for the potential gradient.

A second calculation minimized the uncondensed incremental potential over the plastic-strain increment, plastic multiplier, and two viscous-strain tensors. This provides an independent solution route to the condensed return mapping. For the smooth-cone and apex cases, the maximum normalized potential difference was 6.3838×10^{-16} and the maximum reported state difference was 7.1590×10^{-8}. Full loading–reversal histories also recovered the original elastoplastic implementation in the limit $g_\infty = 1$ and in the infinite-relaxation-time limit with initially zero viscous strains; the maximum stored-state discrepancy was zero.

Fixed-strain relaxation and fixed-stress creep were checked against scalar and matrix-exponential solutions. For the two-branch reference spectrum, the constant-stress recovery times are 25.6110 and 280.2793 s. Table S1 shows temporal refinement of this material-point problem. The stress error is normalized by the reference elastic modulus, and the strain error is dimensionless. The approximately twofold error reduction at each step halving is consistent with backward-Euler integration.

Table S1. Two-branch material-point comparison with exact time evolution.

Number of steps	Step size/s	Relaxation stress error	Creep strain error
30	28.02793	9.8460×10^{-7}	2.5766×10^{-6}
60	14.01397	4.8539×10^{-7}	1.2841×10^{-6}
120	7.00698	2.4087×10^{-7}	6.4085×10^{-7}
240	3.50349	1.1996×10^{-7}	3.2011×10^{-7}

S2. Verification of the wall-diffusion solution

The wall concentration is evaluated using the mixed Dirichlet–Neumann slab solution and Duhamel superposition for piecewise-linear vessel-pressure histories. A short-time image representation complements the Fourier representation. An independent conservative finite-volume discretization used 40, 80, and 160 uniform cells; its semidiscrete evolution was evaluated by a matrix exponential. Table S2 reports maximum errors for a unit normalized concentration step, with $\chi = Dt/h^2$. The fourfold reduction under spatial halving confirms second-order convergence of the independent discretization towards the analytical kernel. An independent numerical convolution of the pressure ramp differed from the implemented Duhamel evaluation by at most 2.1316×10^{-14} MPa.

Table S2. Maximum normalized concentration error of the finite-volume comparison.

Cells	$\chi = 0.01$	$\chi = 0.1$	$\chi = 0.5$
40	2.0493×10^{-3}	1.9877×10^{-4}	8.2515×10^{-5}
80	5.0698×10^{-4}	4.9688×10^{-5}	2.0630×10^{-5}
160	1.2676×10^{-4}	1.2421×10^{-5}	5.1577×10^{-6}

S3. Coupled-solver checks and discretization sensitivity

The macroscopic cylinder uses 96 axisymmetric quadratic elements. The reference pore calculations use curved quadratic hexahedra with coarse and fine partitions of $5 \times 4 \times 4$ and $8 \times 6 \times 6$, respectively, and $3 \times 3 \times 3$ Gauss integration. Spatial comparisons refine the pore mesh while retaining the macroscopic mesh. The temporal comparisons in this section halve waiting and re-pressurization increments from the same saved preparation states; preparation increments retain their original resolution. Sections S6 and S7 extend these checks to a third pore mesh and complete-history time refinement, respectively.

The common fill uses 80 equal pressure increments. The high-pressure hold uses 24 increments with cumulative times logarithmically spaced from 0.1 s to $4\tau_w$. Decompression uses 80 increments, and each axial-pulse leg uses 20 increments. During reference waiting, the nominal step is $\min[600\,\mathrm{s}, \max(0.25\,\mathrm{s}, 0.05(t_w + 4\,\mathrm{s}))]$, shortened to land on requested sampling times. The macro and pore equilibrium solvers require normalized force residuals below 2×10^{-10} and 2×10^{-9}, respectively. The trial yield function F_{tr} is defined in main-text Eq. (12). Candidate recharging increments with $F_{\mathrm{tr}} > 10^{-7}$ MPa trigger an outward-crossing search; Brent refinement uses an absolute tolerance of 10^{-9} in the fractional increment. The reported yield-function residuals are achieved checks, distinct from these prescribed tolerances.

The maximum change in an individual threshold under time refinement was 0.05672 MPa. The largest coarse-to-fine threshold change was 4.30282 MPa for history A after 300 s: 17.25294 versus 12.95012 MPa. Both meshes place B above A at that waiting time, while the absolute pressure intervals differ. Accordingly, history ranking and the classification of a specified pressure are assessed separately in the main text.

For each waiting-stage intervention, define its paired effect as the change in $P_A^* - P_B^*$ relative to the full model under normal re-pressurization. Table S3 reports the largest changes in that effect across the compared times. These checks retain the full prepared state when changing the waiting dynamics.

Table S3. Maximum change in the paired intervention effect, MPa.

Frozen waiting mechanism	Halved time steps	Coarse-to-fine pore mesh
Macroscopic recovery	0.03489	1.75218
Pore-scale recovery	0.01240	0.64214
Pore-gas exchange	0.00686	1.77641

The mechanism dataset contains 198 re-yield pressure evaluations in 16 configurations. Independent reconstruction checked 17,992 accepted records, including repeated shared trajectories. Checks included equilibrium, gas inventory, the equation of state, frozen-variable invariance, unchanged stored energy at intervention switches, and non-negative physical dissipation. The maximum pore-scale energy-balance and inventory-balance residuals were 5.6375×10^{-16} and 4.2758×10^{-18}, both normalized by $E_0 a^3$ because $N = nR_gT$ has energy units. Energy balances use endpoint work and explicitly include backward-Euler numerical dissipation. Their residuals measure discrete consistency; Tables S1–S3 quantify separate discretization effects. The maximum yield-function residual at re-yield was 1.2645×10^{-10} MPa.

S4. Geometric and localization diagnostics

Geometric postprocessing covered 107 saved states from the reference material scenario. Here $\boldsymbol{F} = \boldsymbol{I} + \nabla \boldsymbol{u}$ is the deformation gradient reconstructed from displacement $\boldsymbol{u}$, and $\boldsymbol{I}$ is the identity tensor. The minimum sampled $\det \boldsymbol{F}$ was 0.988668; the maximum small-strain tensor norm was 0.052144; and the maximum cavity-wall displacement divided by the reference radius was 0.020210. Reintegrating cavity volume on the deformed quadratic boundary gave a maximum relative difference of 6.4858×10^{-4} from the linear volume expression. These measures quantify the geometry approximation at the archived states. The outermost element layer retained zero accumulated plastic multiplier throughout this set; the outer radius remained $8a$ in both meshes.

The pore receives the three normal stresses sampled from the macroscopic cylinder. The largest omitted rz shear stress during the reference preparation was 0.01952 MPa. The prescribed reflection symmetry, isotropic constitutive law, and diagonal far-field loading define the local problem. The macroscopic yield and interface-traction checks assess the computed equilibrium branch; geometric buckling requires a separate perturbation or finite-deformation analysis.

For regional yield-margin comparisons, let ρ be distance from the pore center, θ_p the local polar angle from the circumferential axis, and ϕ_p the local azimuth from the radial towards the axial direction. These pore angles are distinct from the macroscopic circumferential coordinate θ. All three regions satisfy $\rho < 2a$: H has $\theta_p < 15°$; M has $\theta_p > 75°$ and $30° < \phi_p < 60°$; Z has $\theta_p > 75°$ and $\phi_p > 75°$. First yielding is searched over all integration points; these subsets describe competing spatial branches.

S5. Transport scales and archived sources

Using the reference transport coefficients from [15], $a^2/D = 4.566$ s, $\tau_p = 181.393$ s, and $h^2/D = 12.6839$ h. Each unloading or axial-pulse leg lasts 1.814 s, whereas re-pressurization lasts 18.139 s. The fixed pore conductance represents local quasi-steady spherical transport. A fixed-boundary spherical concentration-step solution gives an initial conductance multiplier $1 + a/\sqrt{\pi D t}$; its average over 1.814 s is approximately 2.79. This diagnostic motivates the complete-history transient verification in Section S7, which computes the actual coupled threshold correction rather than applying this step multiplier to an evolving boundary history.

The transport coefficients correspond to the 288 K, 70 MPa reference measurements in [15] and are held constant in the present pressure history. The reference spectral shape derives from the recovery study [23]. Their combination defines the material scenario used for mechanism comparisons. Physical validation of absolute thresholds requires measurements under matched material and loading conditions.

Sections S1–S5 use the archived reference calculations. Their source inventory and SHA-256 records are in `manuscript/jes_review/existing_evidence.json`; `extract_existing_evidence.py` regenerates that summary without equilibrium solves. The constitutive records are `screening/viscoelastic_material_audit_round21.json`; wall-diffusion checks are `mechanics/liner_diffusion_bath_round20.json`; geometry records are `screening/viscoelastic_saved_geometry_round21.json`; and macroscopic preparation records are `mechanics/viscoelastic_cylinder_paths_round21.json`. Threshold comparisons use `mechanics/viscoelastic_retention_summary_round21.csv` and the Round22 intervention files indexed by `screening/final_audit_round22.json`. Sections S6–S7 report additional numerical solves and list their own sources. Equation and reference numbers without an S prefix refer to the main article. The figures in Section S9 summarize these checks and the constitutive controls.

S6. Full-history pore-mesh refinement

The additional mesh contains $12 \times 8 \times 8$ curved quadratic hexahedra (768 elements, 20,736 integration points and 20,475 displacement degrees of freedom). It retains the $8a$ outer radius, three-point Gauss integration in each direction, macroscopic mesh, material parameters, pressure program and recharging ramp of the reference calculation. The complete preparation from ambient equilibrium was solved on this

mesh. The common released state was cloned before the two axial-pulse branches; no plastic or viscous fields were interpolated between meshes. The A and B branches were dispatched independently from this same completed common checkpoint.

The waiting path lands on the prescribed waiting times 0, 1, 10, 30, 40, 60, 300, 1800 and 7200 s. Each recharging probe starts from a copied state and therefore does not alter the waiting path. Additional direct probes at 40 and 60 s replay this same path from the original prepared checkpoint; matching coarse- and fine-mesh replays separate mesh effects from slight differences in the original Round21 output schedule. The late states are continued from their own 7200 s pore and cylinder checkpoints, with an intermediate hold node at 18264.84018 s. Coarse- and fine-mesh late controls use the same schedule. Matched scheduling changes the corresponding archived baseline thresholds by at most 0.00148712 MPa.

Table S4. Pressures at the onset of renewed plastic flow for independently prepared pore meshes. All pressures are in MPa. C, F and R denote the $5 \times 4 \times 4$, $8 \times 6 \times 6$ and $12 \times 8 \times 8$ meshes.

Waiting time / s	C: A	C: B	F: A	F: B	R: A	R: B
0	30.1912	19.4404	29.5383	18.7277	29.3006	18.8464
40	24.1594	22.9825	22.1342	22.5286	21.2547	22.5866
60	22.9521	23.4242	20.4189	21.7428	20.0716	20.9201
300	17.2529	17.8772	12.9500	14.4530	13.3329	13.0032
1800	13.4282	14.4188	11.1332	10.8240	10.2958	11.4933
7200	27.8198	25.9182	27.6994	25.2000	26.8798	24.9513
146119	26.2369	21.3652	24.0478	19.3721	22.8826	18.3006

Across the 14 matched thresholds, the maximum absolute successive changes are 4.3028 MPa (C to F) and 1.4498 MPa (F to R). The latter is 10.0311% of the fine-mesh threshold for B at 300 s. Nonmonotone changes occur for B at 0 s, B at 40 s, A at 300 s, B at 1800 s. These are observed discretization differences. Unequal radial/angular refinement and movement of the controlling integration point do not establish an asymptotic Richardson or GCI bound.

The contraction is not uniform: A at 7200 s changes by 0.1205 MPa from C to F, then by 0.8196 MPa from F to R. Its controlling point remains in region M but moves radially from $\rho/a = 1.0580$ to 1.1484 and 1.0946. At 1800 s the refined A maximum also moves angularly, to azimuth 39.375° and polar angle 77.479°, compared with 43.303° and 88.303° on F. The recorded coordinates identify which sampled local maximum controls each threshold; they do not establish that quadrature proximity alone causes the observed shifts.

Table S4 (continued). Sampled ranking signs.

The sampled signs of $P_A^* - P_B^*$ are:

Waiting time / s	C	F	R
0	+	+	+
40	+	-	-
60	-	-	-
300	-	-	+
1800	-	+	-
7200	+	+	+
146119	+	+	+

The 300 s ranking changes on the third mesh; at 1800 s the fine mesh gives the opposite ranking to both other meshes. Accordingly, the computed curves support path-dependent ranking changes, while

the precise crossing times and number of crossings require further resolution. The intersections below are simultaneous finite-mesh operating intervals at the sampled times, not continuum error bounds.

Table S4 (continued). Pressure intersections calculated from unrounded thresholds.

Waiting time / s	Three-mesh A interval / MPa	Three-mesh B interval / MPa	F–R A interval / MPa	F–R B interval / MPa
0	(19.4404, 29.3006)	empty	(18.8464, 29.3006)	empty
40	empty	empty	empty	(22.1342, 22.5286)
60	empty	empty	empty	(20.4189, 20.9201)
300	empty	empty	empty	empty
1800	empty	empty	empty	empty
7200	(25.9182, 26.8798)	empty	(25.2000, 26.8798)	empty
146119	(21.3652, 22.8826)	empty	(19.3721, 22.8826)	empty

On the refined mesh, the whole-volume RMS A–B plastic-strain contrast changes from 0.00019128344 to 0.000183979, retaining 96.18135% of its preparation value at 40.59 h. This is a field-norm comparison on corresponding integration points of the same mesh. Independent reconstruction of all quadratic-element integration weights reproduces the retained fraction to machine precision; the check is archived in `screening/mesh_memory_independent_round23.json`.

The source records are `mechanics/mesh_verification_mesh12x8x8_round23.json`, `mechanics/mesh_baseline_zero_round23.json`, `mechanics/mesh_baseline_early_round23.json` and `mechanics/mesh_baseline_late_round23.json`; the independent extraction and source/checkpoint hashes are in `screening/mesh_verification_audit_round23.json`. The A/B dispatch split changes only job scheduling, with separate outputs merged after completion.

Across 1805 saved accepted records, the maximum normalized discrete energy residual is 5.0087×10^{-15}, pore-inventory residual 2.1971×10^{-16} and normalized force residual 1.8763×10^{-9}. The largest absolute yield-function residual at a refined re-yield pressure is 8.6966×10^{-11} MPa.

S7. Transient spherical exchange: full-history verification

The quasi-steady pore conductance was tested against an infinite-medium spherical diffusion-memory model at the same reference radius. Define $\Delta\pi(t) = \pi_b(t) - p_g(t)$. The local concentration perturbation is superposed on the prescribed wall-diffusion field: the local time derivative of that background is imposed as a source, and the perturbation vanishes at infinity. Thus, the time-varying bath is not treated as a spatially uniform solution of source-free diffusion. For zero initial perturbation, the fixed-radius influx is

$$\dot{N}(t) = \mathcal{L}\left[\Delta\pi(t) + \frac{a}{\sqrt{\pi D}}\int_0^t \frac{\dot{\Delta\pi}(s)}{\sqrt{t-s}}\,\mathrm{d}s\right].$$

Here t is elapsed time from ambient equilibrium, s is the earlier time integrated over the loading history, and $\dot{\Delta\pi} = \mathrm{d}\Delta\pi/\mathrm{d}s$. The inventory $N = nR_gT$, conductance $\mathcal{L}$, reference radius a and diffusivity D retain their definitions in main-text Eq. (9). This solution follows spherical diffusion and Duhamel superposition [28,31]. The conductance, pore radius and transport coefficients are identical to the quasi-steady case. The octant uses one-eighth of the full-sphere conductance and volume. Mechanical pore deformation still changes V in $N = p_gV$, but the diffusion kernel retains the reference radius; moving-boundary diffusion is not introduced.

The convolution is evaluated analytically for the continuous piecewise-linear pressure-difference history. Let t_i and t_{i+1} be successive accepted times. The segment $[t_i, t_{i+1}]$ contributes the following amount to the history integral at $t > t_{i+1}$:

$$\frac{2[\Delta\pi(t_{i+1}) - \Delta\pi(t_i)]}{\sqrt{t - t_i} + \sqrt{t - t_{i+1}}}.$$

This rationalized form avoids cancellation when the long preparation clock precedes the short release and pulse stages. Endpoint backward Euler couples the current segment to pore inventory and volume. The coefficient multiplying the current pressure difference is $1 + 2a/\sqrt{\pi D \Delta t} > 0$, allowing the existing nonlinear pore solve to be reused with an algebraic effective bath and volume offset. The complete convolution history is retained from ambient equilibrium through preparation, waiting and recharge; cloning and fractional first-yield trials do not reset it.

The effective bath is a numerical condensation variable, distinct from π_b. Consequently, the legacy effective-bath chemical-work ledger is used only to check the condensed algebraic solve. No physical total-energy or entropy balance for the transient diffusion field is inferred from that ledger. Pore inventory is checked independently against the actual memory-dependent flux. Across the coarse and fine transient runs, the largest normalized inventory residual is 4.8754×10^{-16} in $E_0 a^3$ units, and the maximum threshold yield-function residual is below 6.3×10^{-11} MPa. Gas pressure, physical bath pressure, pore volume and the algebraic effective bath remain positive.

S7.1. Independent kernel and coupled-ramp checks

Step and ramp kernels agree with their closed forms, and a signed multisegment history agrees with independent quadrature to below 10^{-10}. For a step in pressure difference, the average conductance over a 1.81393 s preparation leg is 2.790286 times the quasi-steady value. This analytical step diagnostic is not applied as a constant multiplier to the coupled history.

An independent fixed-volume benchmark uses a bath ramp of 1 MPa/s from ambient equilibrium. Laplace inversion with the exact spherical kernel gives a pore pressure of 1.58954814 MPa at 18.1393 s. Table S5 shows approximately first-order convergence of the implemented coupled update. Shifting this test by 548000 s changes the 160-step result by only 5.36×10^{-11} MPa. The memory arrays are identical after cloning and remain unchanged during candidate evaluations.

Table S5. Fixed-volume coupled-ramp verification against the independent analytical reference.

Time increments	Computed pore pressure / MPa	Absolute error / MPa
40	1.61824991	0.02870178
80	1.60400132	0.01445318
160	1.59680969	0.00726156
320	1.59319096	0.00364283
640	1.59137373	0.00182559

S7.2. Matched full-history comparison on the coarse mechanical mesh

Both transport models are prepared from ambient equilibrium on the $5 \times 4 \times 4$ mechanical mesh. The base schedule is identical for the two models. Its refined counterpart halves every increment in preparation, waiting and recharge, including the long filling and holding stages and each short release or axial-pulse leg. The common waiting samples are 0, 40, 300, 1800 and 7200 s. The quasi-steady base replay reproduces the archived prescribed-time thresholds within 4×10^{-11} MPa.

Table S6. Coarse-mesh absolute thresholds with complete preparation repeated for each model and time resolution. QS denotes quasi-steady exchange; TR denotes transient spherical exchange; $\Delta t/2$ denotes halving all stages. All pressures are in MPa.

Waiting time / s	History	QS, Δt	TR, Δt	QS, $\Delta t/2$	TR, $\Delta t/2$
0	A	30.191242	30.171271	30.163069	30.142846
0	B	19.440440	19.435524	19.566688	19.561764
40	A	24.159378	24.096919	24.111080	24.047542
40	B	22.982481	22.972283	23.073493	23.063319
300	A	17.252866	17.214844	17.188355	17.149880
300	B	17.877184	17.847427	17.816148	17.786075
1800	A	13.428181	13.462182	13.391086	13.424919
1800	B	14.418764	14.451949	14.376723	14.409763
7200	A	27.819839	27.846617	27.801761	27.828300
7200	B	25.918156	25.922063	25.927825	25.931704

The largest absolute TR-minus-QS correction is 0.062459 MPa at the base resolution and 0.063539 MPa after refinement. The paired correction changes by at most 0.001080 MPa. In contrast, the maximum within-model change under full time refinement is 0.126248 MPa. This quantity is distinct from the 0.05672 MPa check in Section S3, which refined only waiting and recharge from fixed preparation states. The two checks quantify different parts of the time-discretization error.

The refined transient preparation changes the pore inventory relative to QS by +0.0240295% for A and +0.0132969% for B, corresponding to pore-pressure shifts of +0.0081042 and +0.0044780 MPa. The small accumulated inventory correction explains why a large short-time step-kernel multiplier does not produce a similarly large change in the recharging threshold.

S7.3. Fine-mesh confirmation

The transient full preparation and ten recharging tests were repeated on the $8 \times 6 \times 6$ mechanical mesh (Table S7). The QS comparator uses the archived prescribed fine-mesh waiting schedule and the corresponding zero-wait preparation. The largest absolute threshold correction is 0.089942 MPa, for A after 40 s. The maximum relative correction is 0.534815%, for A after 300 s. The five transient A-minus-B differences are +10.791003, -0.474596, -1.518037, +0.309403 and +2.510232 MPa. Every sampled sign agrees with QS, preserving the positive-negative-positive sequence on both tested mechanical meshes.

Table S7. Fine-mesh transport comparison under matched physical histories and recharging ramps. All pressures are in MPa. Differences are evaluated before rounding.

Waiting time / s	QS A	TR A	QS B	TR B	TR A minus B
0	29.538339	29.513939	18.727687	18.722936	+10.791003
40	22.134183	22.044241	22.528620	22.518837	-0.474596
300	12.950016	12.880757	14.452988	14.398794	-1.518037
1800	11.133239	11.151351	10.824013	10.841948	+0.309403
7200	27.699388	27.714208	25.200036	25.203976	+2.510232

The fine-mesh preparation inventory changes are +0.0242777% for A and +0.0137413% for B. These comparisons comprise fifty new threshold evaluations: four model/time-resolution combinations on the

coarse mesh and one transient combination on the fine mesh. They test the short-time exchange approximation for the specified reference histories; they do not supply a universal transport-error bound or establish continuum convergence of the mechanical meshes.

The full result, preparation, waiting and checkpoint files are archived under `mechanics/transient_*_round23*`. `screening/transient_comparison_summary_round23.json` and its CSV companion contain the paired values. `screening/transient_coupled_audit_round23.json` records 49 coupled checks, and `screening/transient_kernel_independent_reference_round23.json` records the independent analytical benchmark. The fixed-radius, infinite-medium and prescribed-background assumptions remain explicit; there is no radial diffusion mesh or finite outer-radius truncation in this verification.

S8. Local pore-wall refinement at fixed physical history

S8.1. Geometry, preparation and observation protocol

The local family starts from the $12 \times 8 \times 8$ quadratic pore mesh, denoted R0. Only its first four radial shells, $1 \leq \rho/a \leq 2$, are subdivided. L1 bisects each original shell and L2 divides it into four; the original outer radial edges and angular nodes remain fixed. The R0 preparation plastic zone extends to the third original shell, $\rho/a = 1.681793$, so the refined zone includes one further shell. All meshes retain the same material, pore radius, outer domain radius, $3 \times 3 \times 3$ quadrature, macroscopic load history and constitutive/threshold tolerances. L1A10 retains L1's radial spacing while increasing each angular division count from 8 to 10.

L3 divides each original near-wall shell into eight, giving $40 \times 8 \times 8 = 2560$ elements. It was added after the L1-to-L2 zero-wait B threshold changed by 0.343938 MPa, exceeding the predeclared target.

Each mesh generates its own complete plastic and viscous preparation from ambient equilibrium. The common fill–hold–release state is shared only between A and B on the same mesh. No state is interpolated from a coarser mesh. Saved waiting states and first-onset probes retain the original physical clocks, pressure histories and gas-exchange law.

At waiting time t_w (s), the nominal time increment is $\max[0.25, 0.05(t_w + 4)]$ s. Each step is capped at 600 s and ends no later than the next prescribed endpoint. Eight additional threshold evaluations use copies of existing accepted states nearest 90, 120, 180, 240, 420, 600, 900 and 1200 s; their actual times are reported in Table S10. These evaluations neither insert an integration endpoint nor alter the parent waiting trajectory. Replaying the original R0 branches from 60 s reproduces the 300/1800 s roots to within 8.54×10^{-11} MPa. All original archive files remain unchanged.

Independent geometry checks establish positive Jacobians, exact parent–child physical mapping in the radial family, unchanged radial-family pore and solid volumes, and affine force/work identities. The small angular-grid change also changes the curved surface approximation: the 8-to-10 pore-volume shift is 4.5624 parts per million, separately recorded from threshold sensitivity. Plastic tangent systems use a checked minimum-degree ordering. Cached constant elastic factors retain the original COLAMD ordering in the earlier dispatchers and use separately verified minimum-degree ordering in later dispatchers. The stage-specific settings are recorded with the source hashes; only factorization ordering changes, with the same matrices, equations, tolerances and accepted physical steps.

Table S8. Local discretizations. The element counts in the radial, polar and azimuthal directions are n_r, n_θ and n_ϕ, respectively. GP denotes a Gauss integration point. Widths and radii are normalized by the pore radius a.

Mesh	Elements	n_r	$n_\theta = n_\phi$	First radial width / a	Minimum GP radius / a
R0	768	12	8	0.189207115	1.021318275
L1	1024	16	8	0.094603558	1.010656356
L2	1536	24	8	0.047301779	1.005325396
L3	2560	40	8	0.023650889	1.002659916

Mesh	Elements	n_r	$n_\theta = n_\phi$	First radial width / a	Minimum GP radius / a
L1A10	1600	16	10	0.094603558	1.010659674

S8.2. Absolute thresholds and the intermediate ranking interval

Tables S9 and S10 report thresholds before rounding their differences. Prescribed waiting states and added accepted-node observations are separated explicitly. The R0 mesh gives positive differences at 300 and 423.758528 s, bounded by negative samples at 234.941205 and 597.899206 s. Local refinement tests this finite interval rather than only its 300 s sign. Opposite-sign neighboring samples bracket an observed crossing; no exact crossing time is inferred from straight connecting lines.

Table S9. Re-yield pressures at the prescribed waiting times on each local mesh. All pressures are in MPa.

Waiting time / s	Mesh	A / MPa	B / MPa	A minus B / MPa
0.000000	R0	29.300637	18.846397	+10.454240
0.000000	L1	29.172591	18.447884	+10.724707
0.000000	L2	29.130879	18.103946	+11.026933
0.000000	L3	29.124324	17.913861	+11.210463
0.000000	L1A10	29.157892	18.503329	+10.654563
40.000000	R0	21.254684	22.586558	-1.331874
40.000000	L1	21.643540	22.138327	-0.494787
40.000000	L2	21.431948	22.034055	-0.602107
40.000000	L3	21.417802	21.925960	-0.508159
40.000000	L1A10	21.333818	22.243208	-0.909390
60.000000	R0	20.071636	20.920065	-0.848429
60.000000	L1	20.033978	20.520536	-0.486558
60.000000	L2	20.051061	20.776391	-0.725330
60.000000	L3	19.953284	20.408352	-0.455069
60.000000	L1A10	19.998256	20.574793	-0.576537
300.000000	R0	13.332929	13.003194	+0.329735
300.000000	L1	13.402635	13.202906	+0.199729
300.000000	L2	12.963974	13.359568	-0.395595
300.000000	L3	12.877417	13.100079	-0.222663
300.000000	L1A10	13.303130	13.553123	-0.249993
1800.000000	R0	10.295781	11.493300	-1.197519
1800.000000	L1	10.812330	10.607514	+0.204817
1800.000000	L2	10.716610	10.692256	+0.024354
1800.000000	L3	10.592257	10.517045	+0.075212
1800.000000	L1A10	10.225233	10.872044	-0.646812
7200.000000	R0	26.879756	24.951273	+1.928483
7200.000000	L1	26.720114	24.624783	+2.095331
7200.000000	L2	26.874499	24.449183	+2.425316
7200.000000	L3	26.861045	24.354228	+2.506817
7200.000000	L1A10	26.750832	24.609017	+2.141815

Waiting time / s	Mesh	A / MPa	B / MPa	A minus B / MPa
146118.721460	R0	22.882576	18.300553	+4.582022
146118.721460	L1	21.945721	17.198946	+4.746775
146118.721460	L2	21.432000	16.639509	+4.792491
146118.721460	L3	21.180272	16.368780	+4.811492
146118.721460	L1A10	21.942160	17.196083	+4.746077

Table S10. Additional evaluations at unchanged accepted waiting nodes. All pressures are in MPa.

Waiting time / s	Mesh	A / MPa	B / MPa	A minus B / MPa
90.557148	R0	18.306731	19.091178	-0.784447
90.557148	L1	18.398766	19.390778	-0.992011
90.557148	L2	18.325167	18.981488	-0.656320
90.557148	L3	18.221633	19.042887	-0.821255
90.557148	L1A10	18.340329	19.248304	-0.907974
122.715622	R0	16.878628	18.370876	-1.492248
122.715622	L1	17.066968	18.069832	-1.002865
122.715622	L2	16.900138	18.025928	-1.125790
122.715622	L3	16.774125	17.918915	-1.144790
122.715622	L1A10	17.013760	18.029594	-1.015835
183.216686	R0	14.609135	16.206539	-1.597404
183.216686	L1	14.792987	16.043477	-1.250490
183.216686	L2	14.932939	15.838609	-0.905670
183.216686	L3	14.836877	15.733703	-0.896826
183.216686	L1A10	15.115134	15.951723	-0.836589
234.941205	R0	13.917488	14.671076	-0.753588
234.941205	L1	14.079039	14.525141	-0.446102
234.941205	L2	13.667887	14.280423	-0.612536
234.941205	L3	13.702728	14.164101	-0.461373
234.941205	L1A10	14.442723	14.512946	-0.070223
423.758528	R0	12.611391	12.287931	+0.323461
423.758528	L1	11.635448	12.470135	-0.834687
423.758528	L2	11.735523	11.974779	-0.239256
423.758528	L3	11.673204	11.900057	-0.226853
423.758528	L1A10	11.919719	12.593571	-0.673852
597.899206	R0	11.583581	11.809067	-0.225485
597.899206	L1	11.180040	10.944058	+0.235982
597.899206	L2	11.007109	11.036257	-0.029149
597.899206	L3	11.068298	11.155541	-0.087242
597.899206	L1A10	11.191479	11.137843	+0.053636
885.279259	R0	10.562454	11.512721	-0.950267
885.279259	L1	10.911009	10.603814	+0.307195
885.279259	L2	10.720016	10.692885	+0.027132

Waiting time / s	Mesh	A / MPa	B / MPa	A minus B / MPa
885.279259	L3	10.591948	10.606702	-0.014754
885.279259	L1A10	10.208157	10.872006	-0.663849
1187.719258	R0	10.224039	11.422477	-1.198438
1187.719258	L1	10.743531	10.524125	+0.219406
1187.719258	L2	10.637667	10.610241	+0.027426
1187.719258	L3	10.508613	10.423660	+0.084952
1187.719258	L1A10	10.124384	10.791808	-0.667423

Across R0, L1, L2, L3, L1A10, the following waiting times have a common favored history and pass both the last radial and angular criteria for numerical resolution of the ranking. A is favored at 0, 7200, 146118.721 s. B is favored at 40, 90.557, 122.716, 183.217 s. This list reports sampled waiting times; it does not estimate the times of crossings between them.

S8.3. Practical resolution and pressure-window convergence

A 0.1 MPa numerical-resolution target was declared before examining the local-refinement thresholds, equal to one tenth of the 1 MPa distinguishability scale. Each comparison reports the maximum individual A and B threshold changes and the maximum A-minus-B change over all 15 matched waiting times. Passing this target requires all three quantities to be at most 0.1 MPa. It is an empirical finite-mesh criterion, not a Richardson estimate, a continuum error bound or an uncertainty in material properties. Radial and angular comparisons remain separate.

At an individual waiting time, the ranking is treated as numerically resolved only if the two grids give the same sign of ΔP^* and its absolute value on the finer grid exceeds both 0.1 MPa and twice its change between the two grids. Pressure-window endpoints are the two individual thresholds, so their convergence also requires absolute-threshold stability. The existence of an intersection is recorded separately from its endpoint convergence. If the ranking changes, the identity of the history with the higher threshold must be retained when comparing intervals.

Table S11. Successive radial and separate angular changes over matched prescribed and additional waiting times. All pressure changes are absolute values in MPa.

Comparison	Matched samples	Max change in A	Max change in B	Max change in A minus B	0.1 MPa target
R0 to L1	15	0.975944	1.101607	1.417843	no
L1 to L2	15	0.513721	0.559437	0.595431	no
L2 to L3	15	0.251728	0.368038	0.270261	no
L1 to L1A10	15	0.702852	0.350217	0.971044	no

The radial intersection uses L2, L3. The intersection across the local radial and angular grids uses L1, L2, L3 and L1A10. These are the explicitly stated comparison sets in Table S12.

Table S12. Shared open pressure intervals at prescribed waiting times. Empty intersections have no favored-history label or endpoints and are assigned zero width. All pressures are in MPa.

Waiting time / s	Mesh set	Favored history	Lower / MPa	Upper / MPa	Width / MPa
0.000000	radial	A	18.103946	29.124324	11.020378

Waiting time / s	Mesh set	Favored history	Lower / MPa	Upper / MPa	Width / MPa
0.000000	local+angular	A	18.503329	29.124324	10.620995
40.000000	radial	B	21.431948	21.925960	0.494012
40.000000	local+angular	B	21.643540	21.925960	0.282421
60.000000	radial	B	20.051061	20.408352	0.357291
60.000000	local+angular	B	20.051061	20.408352	0.357291
300.000000	radial	B	12.963974	13.100079	0.136106
300.000000	local+angular	—	—	—	0.000000
1800.000000	radial	—	—	—	0.000000
1800.000000	local+angular	—	—	—	0.000000
7200.000000	radial	A	24.449183	26.861045	2.411862
7200.000000	local+angular	A	24.624783	26.720114	2.095331
146118.721460	radial	A	16.639509	21.180272	4.540763
146118.721460	local+angular	A	17.198946	21.180272	3.981326

The accompanying file `local_window_convergence_round24.csv` reports pairwise endpoint and width shifts at all 15 waiting times, including changes of the favored history. Signed shifts use the newer grid minus its stated parent. The predeclared window criterion requires the same favored history and lower and upper endpoint changes of at most 0.1 MPa. A separate, stricter indicator requires the width to agree to 0.1 MPa as well. The criterion for numerical resolution of the ranking is tabulated independently.

S8.4. Critical locations and source traceability

Table S13 locates the point at which plastic flow resumes at 300 s and at the largest *sampled* difference within $60 < t_w < 1800$ s on each mesh. Duplicate times are listed once. A largest sampled value need not be positive or be the maximum of the continuous-time response. All locations are actual integration points from the threshold root; no stress or plastic field is interpolated between meshes. Region labels use the same H/M/Z definitions as the main text, while the threshold search itself includes every integration point.

Table S13. Actual critical integration points at the intermediate diagnostic times.

Waiting time / s	Mesh	History	ρ/a	Polar / deg	Azimuth / deg	Region
300.000000	R0	A	1.021318	88.729309	43.729313	M
300.000000	R0	B	1.021318	88.729309	43.729313	M
300.000000	L1	A	1.047296	88.729309	43.729313	M
300.000000	L1	B	1.010656	88.729309	43.729313	M
885.279259	L1	A	1.047296	88.729309	43.729313	M
885.279259	L1	B	1.047296	88.729309	43.729313	M
300.000000	L2	A	1.023645	88.729309	43.729313	M
300.000000	L2	B	1.005325	88.729309	43.729313	M
1187.719258	L2	A	1.052627	88.729309	43.729313	M
1187.719258	L2	B	1.041965	88.729309	43.729313	M
300.000000	L3	A	1.035471	88.729309	43.729313	M
300.000000	L3	B	1.011820	88.729309	43.729313	M

Waiting time / s	Mesh	History	ρ/a	Polar / deg	Azimuth / deg	Region
1187.719258	L3	A	1.059121	88.729309	43.729313	M
1187.719258	L3	B	1.049961	88.729309	43.729313	M
300.000000	L1A10	A	1.047299	88.984252	43.984254	M
300.000000	L1A10	B	1.010660	88.984252	43.984254	M
597.899206	L1A10	A	1.010661	79.984254	40.500000	M
597.899206	L1A10	B	1.047299	88.984252	43.984254	M

The source extraction is `screening/local_refinement_summary_round24.json`; the independent geometry record, predeclared assessment and publication audit are `screening/local_geometry_independent_round24.json`, `screening/local_refinement_predeclared_round24.json` and `screening/audit_round24_publication.json`. Full preparation, held-state, observation and root checkpoints are archived under `mechanics/local_*round24*`. The new observations, thresholds, critical locations and figure arrays carry their source hashes. The reference mechanism interventions and transient-transport calculations retain their original mesh scope; they have not been relabeled as locally refined interventions.

S8.5. Integration-point sampling of a virgin spherical cavity

A virgin elastic spherical annulus of inner radius a and outer radius $b = 8a$ starts under a uniform internal and external pressure $P_0 = 0.1$ MPa. The internal pressure increases by Δp while the outer pressure remains fixed. Under these tractions, Lamé's solution gives the von Mises stress increment per unit pressure as $3a^3b^3/[2(b^3 - a^3)\rho^3]$ and the corresponding stress-trace increment as $3a^3/(b^3 - a^3)$. For the virgin Drucker–Prager surface, $q_\sigma + \alpha \operatorname{tr} \boldsymbol{\sigma} = \sigma_T(1+\alpha)$, the pressure increment at first yield at radius ρ is

$$\Delta p_y(\rho) = \frac{\sigma_T(1+\alpha) + 3\alpha P_0}{\dfrac{3a^3b^3}{2(b^3-a^3)\rho^3} + \dfrac{3\alpha a^3}{b^3-a^3}}, \qquad \alpha = \frac{1.16-1}{1.16+1}, \qquad \sigma_T = 36 \text{ MPa}.$$

Here ρ is distance from the pore center, q_σ is the von Mises equivalent stress, σ_T is the initial tensile yield strength, and α is the pressure-sensitivity coefficient from main-text Eq. (4). The scalar b denotes the outer radius only in this benchmark. The exact threshold increases with radius and reaches its minimum at the wall. Evaluating the same solution at the actual integration-point coordinates isolates the effect of sampling inside the material. A separate equilibrium solve uses the original element mapping, $3\times3\times3$ integration rule and pressure traction on each mesh. Its difference from the analytic threshold over the sampled points includes the effects of the finite-element (FE) stress field, geometric approximation and selection of the controlling point. The absolute internal pressure equals $P_0 + \Delta p_y$.

Table S14. Virgin spherical-cavity yield increments and sources of threshold shift. All pressures and shifts are in MPa. The sampling shift is the exact integration-point threshold minus the exact wall threshold. The final column compares the computed and exact integration-point thresholds.

Mesh	Exact wall increment	Exact GP-set increment	FE GP-set increment	Sampling shift	FE minus exact GP
R0	25.734770	27.415451	27.565522	1.680681	0.150071
L1	25.734770	26.566037	26.607960	0.831267	0.041923
L2	25.734770	26.147985	26.159539	0.413215	0.011554
L3	25.734770	25.940614	25.944010	0.205844	0.003397

Mesh	Exact wall increment	Exact GP-set increment	FE GP-set increment	Sampling shift	FE minus exact GP
L1A10	25.734770	26.566298	26.608078	0.831528	0.041779

On L3 the computed integration-point threshold exceeds the exact wall threshold by 0.209240 MPa. Sampling accounts for 0.205844 MPa, or 98.38% of this shift. Increasing the angular resolution from L1 to L1A10 changes the computed threshold by 0.000118 MPa. This comparison isolates one source of sensitivity in the first-yield pressure under a prescribed elastic loading. The A and B thresholds in Tables S9–S10 come from the separate history-dependent calculations.

The source record is `screening/lame_yield_sampling_round24.json`. It gives source hashes, equilibrium residuals, first-yield residuals, and the radii and indices of the controlling integration points. The largest relative free-force residual is 1.16×10^{-13} and the largest absolute FE yield-function residual at the computed threshold is 7.11×10^{-15} MPa.

S9. Supplementary figures, constitutive controls and pressure-window accuracy

S9.1. Independent solution checks

Figure S1 summarizes the independent time-integration and wall-diffusion comparisons in Sections S1 and S2. The constitutive tangent, incremental potential and coupled balances are checked separately in Sections S1 and S3. These comparisons test numerical accuracy against independently obtained solutions; the material inputs are the reference parameter set in main-text Table 1.

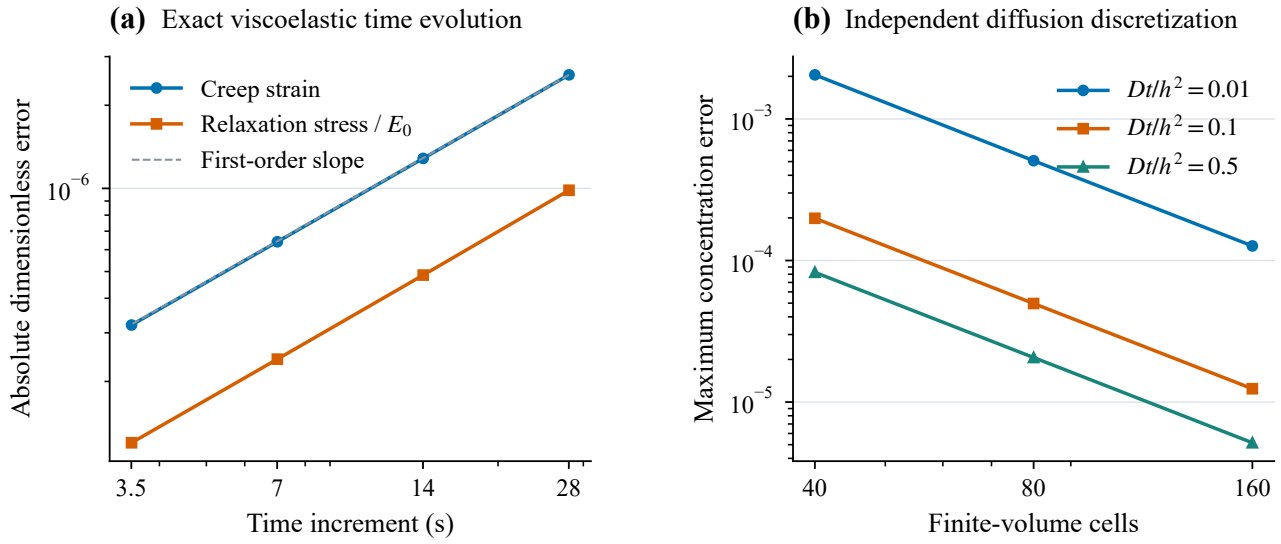


Figure S1: Independent numerical verification. (a) Errors in two-branch creep and relaxation relative to matrix-exponential solutions as the time step decreases; stress error is normalized by the instantaneous modulus E_0. The dashed line shows first-order scaling. (b) Maximum concentration differences between the analytical slab solution and conservative finite-volume solutions at dimensionless times Dt/h^2, where D is diffusivity, t is elapsed time and h is liner thickness. See Tables S1 and S2.

S9.2. Constitutive controls and numerical sensitivity of mechanism comparisons

The elastic–plastic limit and an alternative recovery spectrum repeat the full preparation and waiting protocol on the coarse pore mesh. The elastic–plastic limit uses $g_\infty = 1$. The alternative spectrum

uses $g_\infty = 0.468$, $(g_1, g_2) = (0.409, 0.123)$ and $(\tau_1, \tau_2) = (21.48, 177.36)$ s, where g_∞ is the equilibrium stiffness fraction, g_i is the stiffness fraction of Maxwell branch i, and τ_i is its relaxation time. Each control generates its own plastic and viscous preparation state. In contrast, the waiting-stage mechanism interventions in the main article preserve the preparation state.

Table S15. Threshold differences for independently prepared constitutive controls. The entries are $\Delta P^* = P_A^* - P_B^*$ in MPa on the coarse pore mesh.

Waiting time	Reference spectrum	Alternative spectrum	Elastic–plastic limit
0	+10.751	+7.199	+17.902
5 min	-0.624	+0.407	+17.578
2 h	+1.902	+0.742	+15.873
40.59 h	+4.872	+2.891	+11.781

At 5 min, the reference spectrum favors B, whereas the two controls favor A. At 40.59 h, the differences span 2.891–11.781 MPa. Recovery properties influence both the state formed during preparation and its subsequent evolution. The four common sampling times establish these differences between constitutive models; they do not locate every ranking change in the two control trajectories.

Figure S2b reports changes in the effect of freezing one mechanism, where the effect is the change in ΔP^* relative to the calculation with all mechanisms active at the same waiting time. Halving waiting and recharging increments from a fixed preparation changes individual thresholds by at most 0.05672 MPa and a paired intervention effect by at most 0.03489 MPa. The largest change in an intervention effect between the coarse and fine pore meshes is 1.77641 MPa (Table S3). The separate spatial and temporal comparisons retain their original preparation and mesh definitions.

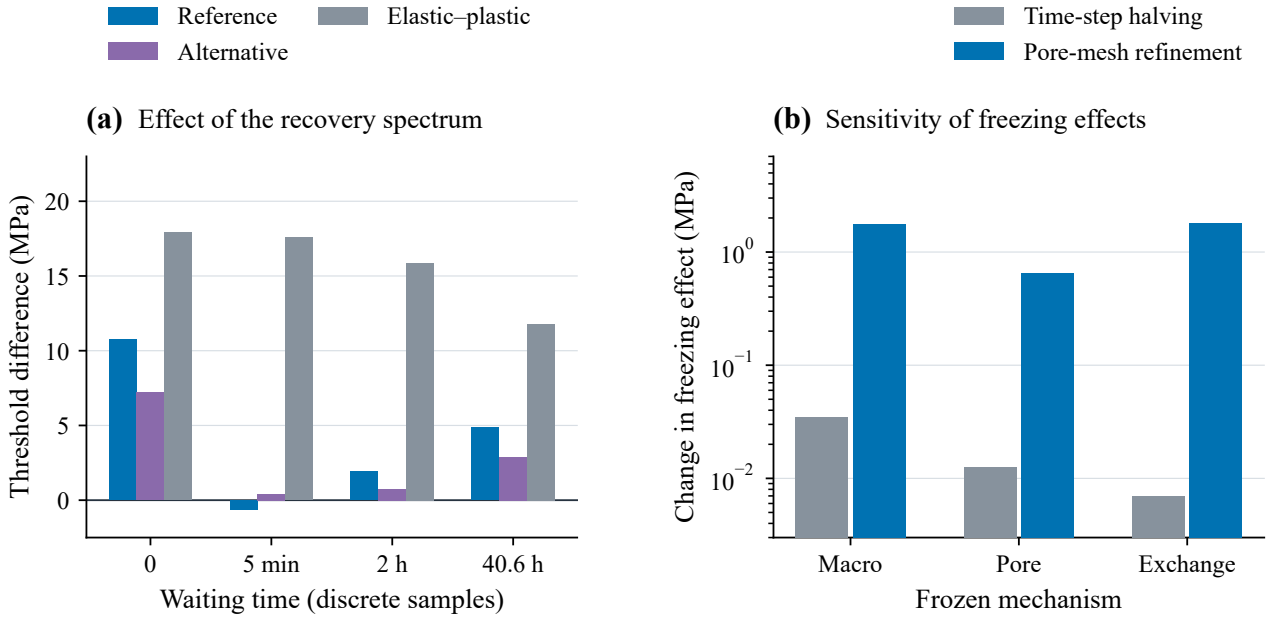


Figure S2: Constitutive controls and numerical sensitivity. (a) Threshold differences for the reference spectrum, alternative spectrum and elastic–plastic limit, each prepared independently on the coarse pore mesh. (b) Maximum change across matched waiting times in the effect of freezing one mechanism on ΔP^* when time steps are halved or the pore mesh is refined. The vertical axis in (b) is logarithmic. Values are given in Tables S3 and S15.

S9.3. Pore-wall refinement and additional changes in ranking

Figure S3 compares absolute thresholds on the global meshes C, F and R0 and on the local radial and angular refinements defined in Section S8. The largest change from F to R0 is 1.4498 MPa for B at 300 s. At the same wait, ΔP^* is +0.330 MPa on R0, +0.200 MPa on L1, -0.396 MPa on L2, -0.223 MPa on L3 and -0.250 MPa on L1A10. Figure S4 adds observations on either side of 300 s at the original accepted waiting nodes. L2 and L3 contain no positive interval bounded by negative observations over 60–1800 s. L1 and L1A10 contain positive sampled values within such bounds, reaching +0.200 MPa at 300 s and +0.054 MPa at 597.899 s, respectively, but none meets the resolution criterion in Section S8.3. These additional changes of sign are therefore excluded from the main mechanism interpretation.

The principal A–B–A sequence is supported at 0, 122.716 and 7200 s on R0 and all four refinements. At each of these times, the advantage exceeds 1 MPa and satisfies the last radial and angular comparison criteria. In contrast, the common negative sign at 60 s does not pass the last radial criterion: the L3 advantage is 0.455 MPa, less than twice its change from L2 (0.541 MPa). The resolved stages are distinguished from these small, mesh-sensitive differences.

Pointwise onset is more sensitive than integrated plastic-state measures. At 300 s, A's controlling radii ρ/a are 1.021318, 1.047296 and 1.023645 on R0, L1 and L2; B selects the nearest-wall integration point on those grids. On L3, the corresponding radii are 1.035471 for A and 1.011820 for B. Neither is the nearest available point, at 1.002660 (Table S13). Over $1 \leq \rho/a \leq 2$, the L1-to-L2 mean accumulated plastic multipliers change by at most 0.0342%, and the root-mean-square A–B plastic-strain difference changes by 0.0607%. The corresponding L2-to-L3 changes are at most 0.0080% and +0.0144%, while B's onset pressure changes by -0.259 MPa. The independent elastic-cavity calculation in Section S8.5 isolates the contribution of off-wall integration-point sampling without applying a correction to these history-dependent thresholds.

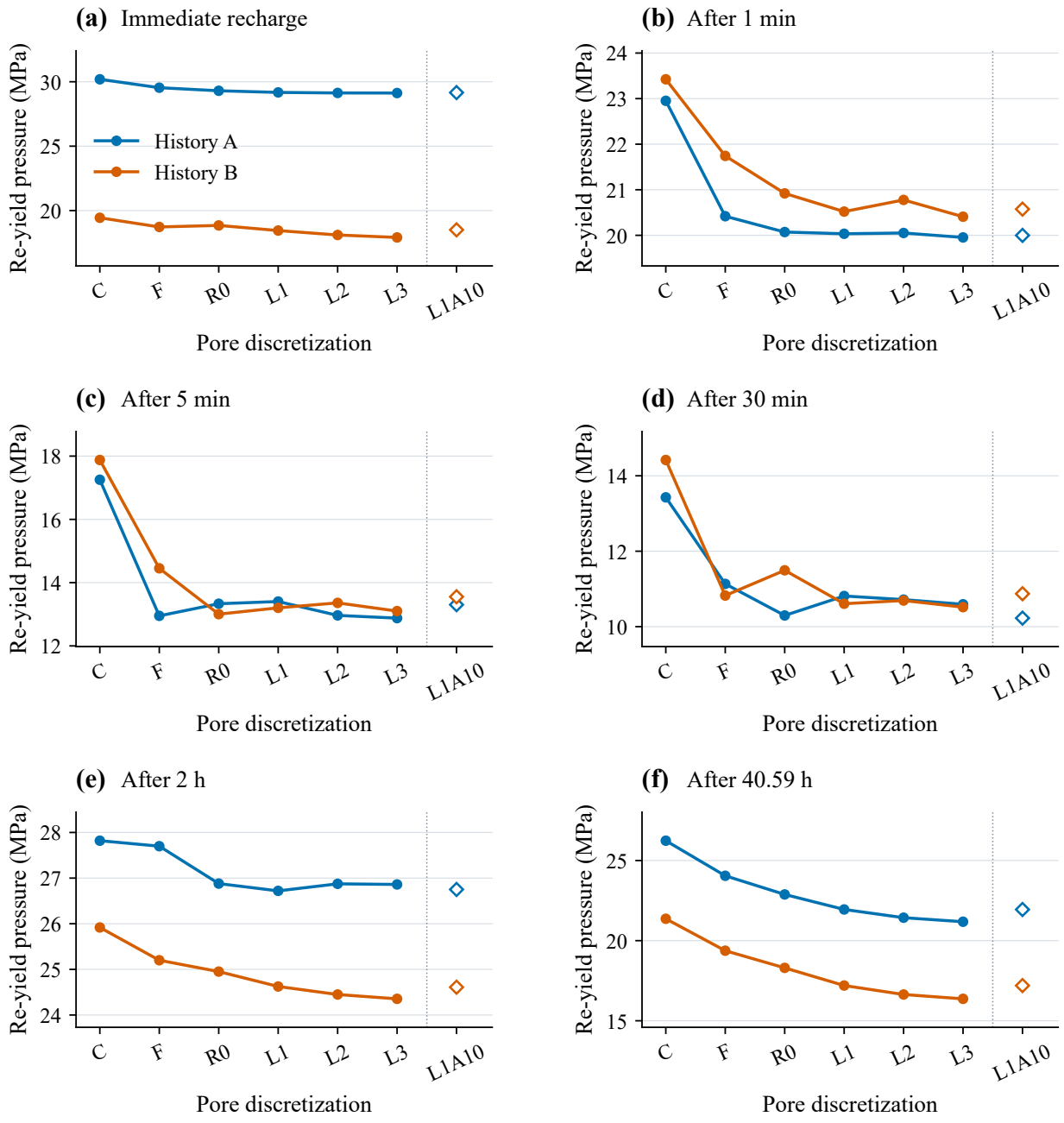
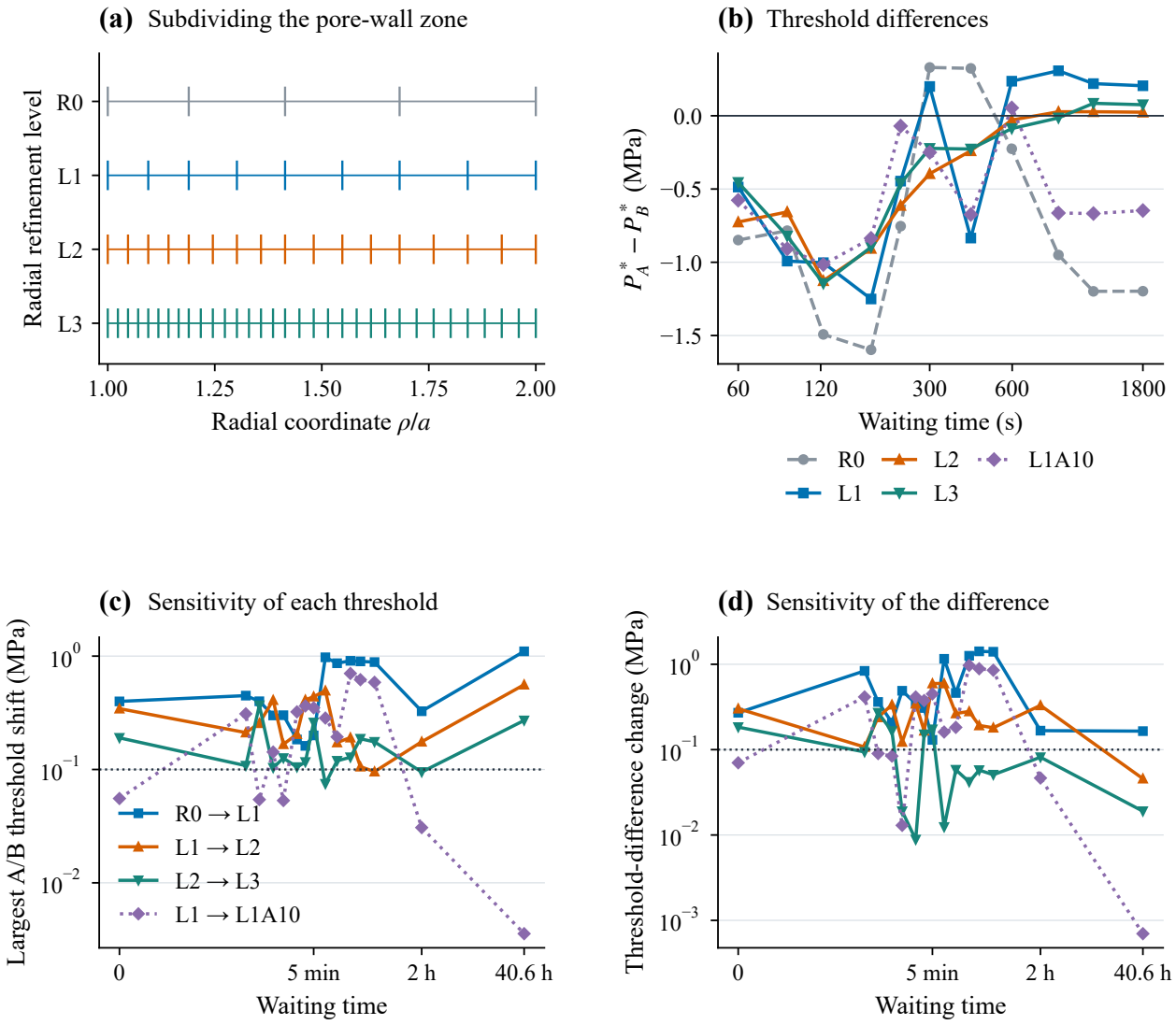


Figure S3: Absolute pressures at the onset of renewed plastic flow under global and local pore refinement. (a) Immediately after preparation; (b) after 1 min; (c) after 5 min; (d) after 30 min; (e) after 2 h; (f) after 40.59 h. C, F and R0 contain 80, 288 and 768 elements. The local radial meshes and separate angular control are defined in Table S8. Every mesh repeats the full preparation. All cases retain the quasi-steady exchange law, macroscopic mesh and outer pore-domain radius $8a$. Lines join computed values.



Dotted horizontal lines: the declared 0.1 MPa numerical-resolution target.

Markers are computed points; connecting lines do not locate exact crossings.

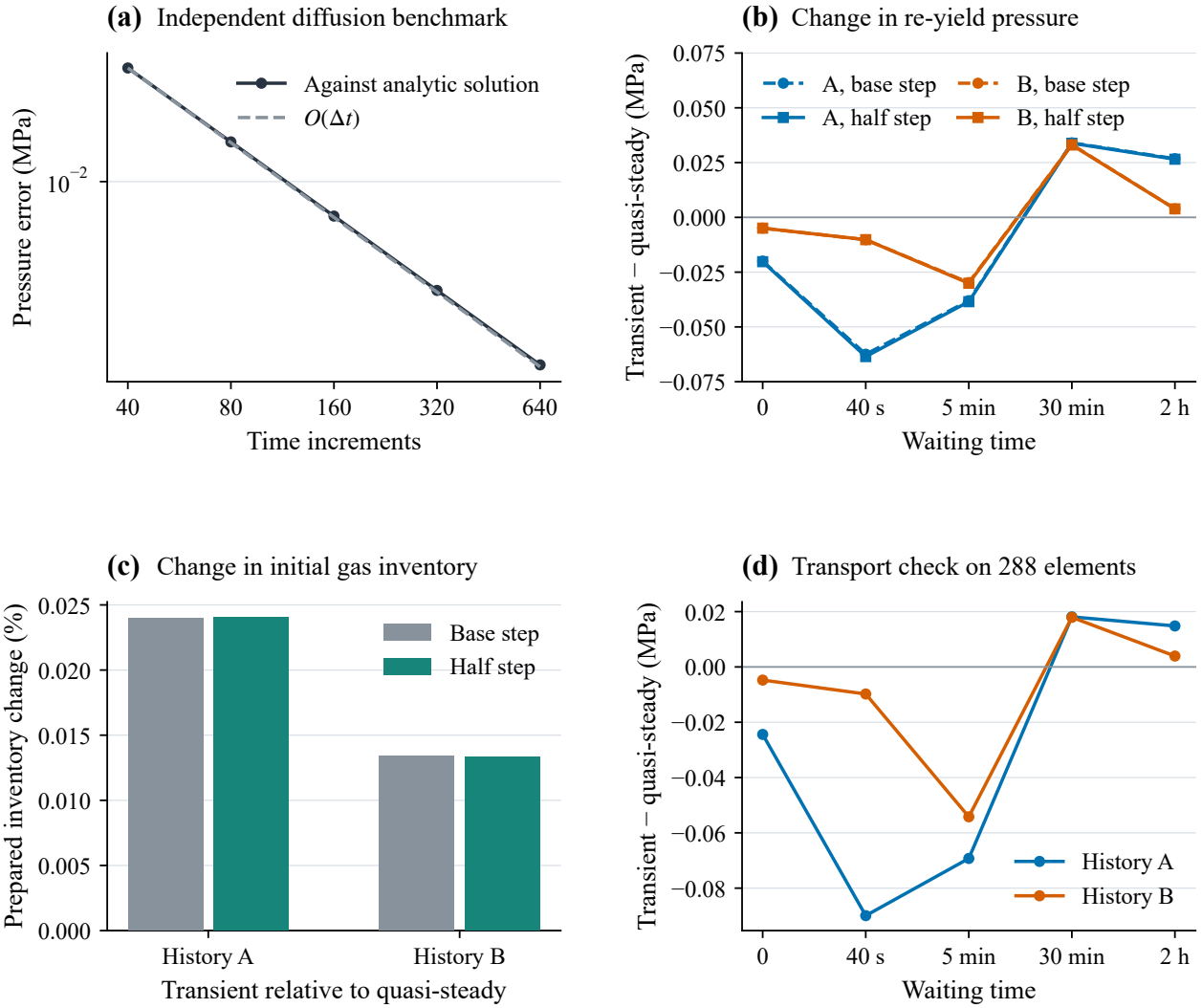
Figure S4: Local pore-wall refinement. (a) Radial element boundaries over $1 \leq \rho/a \leq 2$. (b) A-minus-B pressure differences at prescribed and additional accepted waiting times. (c) Maximum change in either individual threshold at each matched time. (d) Change in the threshold difference. The angular control changes angular divisions at fixed radial edges. Dotted horizontal lines in (c,d) mark the 0.1 MPa numerical target. Points are computed observations; connecting lines do not determine exact crossing times. The external loading history and waiting-step schedule are unchanged.

S9.4. Transient spherical exchange

Figure S5 summarizes the full-history comparison in Section S7. The fixed-volume benchmark approaches the independent Laplace solution with approximately first-order time accuracy. On the coarse mesh, transient exchange changes a threshold by at most 0.062459 MPa at the base resolution and 0.063539 MPa when every time step is halved. The paired correction changes by only 0.001080 MPa, while the largest absolute threshold change under that full-history refinement is 0.126248 MPa. This latter check includes preparation, unlike the fixed-preparation time refinement in Section S3.

The maximum transient correction on the fine mesh is 0.089942 MPa. All five sampled ranking signs are preserved. The prepared pore inventory changes by +0.0242777% for A and +0.0137413% for B on

the fine mesh, consistent with the small pressure correction. These results test the quasi-steady exchange approximation for the prescribed histories. The mechanism-freezing comparisons retain the quasi-steady reference law and their original coarse or fine mechanical meshes.



Full preparation is repeated for each transport model and time resolution.

Figure S5: Verification of transient spherical exchange. (a) Pressure error for a fixed-volume coupled ramp relative to an independent Laplace solution; the dashed line shows first-order time scaling. (b) Transient-minus-quasi-steady threshold corrections on the 80-element pore mesh at base and halved time steps. (c) Relative changes in prepared gas inventory for those calculations. (d) Threshold corrections on the 288-element mesh. Every model and time resolution repeats preparation, waiting and recharging. Tables S5–S7 give the numerical values.

S9.5. Absolute thresholds and pressure-window endpoints

The largest L2-to-L3 changes over the 15 matched waiting times are 0.251728 MPa for A, 0.368038 MPa for B and 0.270261 MPa for ΔP^*. The L1-to-L1A10 angular comparison gives 0.702852, 0.350217 and 0.971044 MPa, respectively (Table S11). Neither comparison meets the 0.1 MPa target at every time. The endpoint criterion in Section S8.3 is met at two times for the last radial pair and five times for the angular pair; adding the width criterion leaves these counts unchanged.

At 2 h, both comparisons meet the endpoint and width criteria. The last radial changes are -0.095 and -0.013 MPa for the lower and upper endpoints, and +0.082 MPa for width. The shared interval for

L1, L2, L3 and L1A10 is (24.625, 26.720) MPa. At 40.59 h, however, the same radial refinement shifts the lower and upper endpoints by -0.271 and -0.252 MPa while changing width by only +0.019 MPa. An almost unchanged width can therefore accompany a translated pressure interval.

All seven global and local meshes favor A at 40.59 h, but they have no common distinguishing terminal pressure: the highest B threshold is 21.365 MPa on C and the lowest A threshold is 21.180 MPa on L3. The intervals in main-text Figure 8 use the stated four locally refined meshes. Their initial and late intersections are (18.503, 29.124) and (17.199, 21.180) MPa, respectively, so a 20 MPa terminal pressure makes only B yield at both waits. At 2 h, the same pressure is below both thresholds. Table S12 and `local_window_convergence_round24.csv` report the separate interval-existence and endpoint-convergence results.